Controlled Packet Capture and 802.11 Protocol Analysis Lab - Lab Setup

How can I perform packet captures and analysis without pcaps(Packet Captures)? This is a question I often receive and for multiple reasons. Some of those reasons include.

- Hey, I am trying to learn 80211 but don't have access to pcaps or access points
- I am studying for an exam and need to test and learn via protocol analysis
- I kinda understand the 4-way handshake but not really!
- I don't want to use live pcaps and would rather a lab environment.
- I want to test this feature and understand it prior to attempting in production.
- I don't have access to any access points (APs) to test with.
- I want to learn EAP and troubleshoot authentication methods.
- I want to learn how to identify retries.

Honestly, the list goes on. I think we get the point.

To that end, I decided to create a series of blogs post to assist with guidance from Lab setup to analysis. Our goal is to create a repeatable, low impact, low cost wireless Lab environment that can be used for learning and troubleshooting purpose. To accomplish this goal, we will utilize free, readily accessible software, and external wireless LAN Adapters.

I strongly believe in building your own Lab vs pre-configured Lab environments. I feel the learning benefits from figuring out what needs to be built is critical to understanding. I have taken this approach my whole career and specifically while preparing for the CCIE Lab exam. I strongly believe that without creating labs and learning from trial and error, even with the amount of experience I had at the time, I may not have passed the CCIE Lab. To this day, I encourage anyone willing to learn to lab, lab, lab, and then lab some more. In the end, you will learn to love building labs.

If you can think it, you can lab it!

Lab Requirements

This first post will walk through identifying the required hardware, software, and establishing our base Lab environment.

Subsequent labs in this series, will build off this foundation.

Prerequisites

- Linux system with root access. My preference. (Linux Mint)(Wilma)(22)
- Wireless adapter that supports AP and Monitor Mode.(Panda Wireless PAU0F AXE3000)
- Preferably a second adapter that supports monitor mode for clean over-the-air captures

- Packages:
  - **hostapd** - A Linux daemon that turns a wireless network interface into a Wi-Fi access point, managing authentication, encryption, and client connections.
  - **dnsmasq** - A lightweight network service that provides DNS forwarding, DHCP address assignment, and network configuration for connected devices.
  - **iw** - A command-line utility for configuring and monitoring wireless network interfaces on Linux, including Wi-Fi channels, frequencies, and connection status.
  - **wireshark** - A network protocol analyzer used to capture, inspect, and troubleshoot network traffic, including wireless communications.
  - **aircrack-ng** - A suite of wireless network security tools used to assess Wi-Fi security, monitor wireless traffic, capture packets, and evaluate the strength of Wi-Fi authentication and encryption.

Lab Setup

Now that we have our roadmap and understanding of the requirements. Let's Build a Lab!

First step, let's open a terminal and update and install our packages. Answer Yes to install.

Install Required Packages

```
sudo apt update
sudo apt install hostapd dnsmasq iw wireshark aircrack-ng
```

Identify Wireless LAN Adapaters

Next, Let's identify the Wlan adapters on our system.

```
iwconfig
```

We can now identify the adapters on this system. The adapter ending in "950a" will be used for Hostapd. The adapter ending in "3065" will be placed in monitor mode and used packet capture.

```
wlx9cefd5f6950a  IEEE 802.11  ESSID:off/any
          Mode:Managed  Access Point: Not-Associated   Tx-Power=3 dBm
          Retry short limit:7   RTS thr:off   Fragment thr:off
          Power Management:on

wlx9cefd5f63065  IEEE 802.11  ESSID:off/any
          Mode:Managed  Access Point: Not-Associated   Tx-Power=3 dBm
          Retry short limit:7   RTS thr:off   Fragment thr:off
          Power Management:on
```

As a side note, you can issue the following to review and identify the capabilities of each adapter.

```
iw phy
```

One last important piece of information we need to check is the Regulatory Domain. Frequency availability is a Regulatory restrictions. Let's make sure we are set to the proper Domain.

We are in the US, as such we will set the regulatory Domain to US

```
iw reg get
```

Output.
Notice how the global is not set. Let's set it to US so that we can use all the frequencies available in US.

```
global
country 00: DFS-UNSET
        (755 - 928 @ 2), (N/A, 20), (N/A), PASSIVE-SCAN
        (2402 - 2472 @ 40), (N/A, 20), (N/A)
        (2457 - 2482 @ 20), (N/A, 20), (N/A), AUTO-BW, PASSIVE-SCAN
        (2474 - 2494 @ 20), (N/A, 20), (N/A), NO-OFDM, PASSIVE-SCAN
        (5170 - 5250 @ 80), (N/A, 20), (N/A), AUTO-BW, PASSIVE-SCAN
        (5250 - 5330 @ 80), (N/A, 20), (0 ms), DFS, AUTO-BW, PASSIVE-SCAN
        (5490 - 5730 @ 160), (N/A, 20), (0 ms), DFS, PASSIVE-SCAN
        (5735 - 5835 @ 80), (N/A, 20), (N/A), PASSIVE-SCAN
        (57240 - 63720 @ 2160), (N/A, 0), (N/A)
```

Set Regulatory Domain

```
sudo iw reg set US
```

Verify

```
global
country US: DFS-FCC
        (902 - 904 @ 2), (N/A, 30), (N/A)
        (904 - 920 @ 16), (N/A, 30), (N/A)
        (920 - 928 @ 8), (N/A, 30), (N/A)
        (2400 - 2472 @ 40), (N/A, 30), (N/A)
        (5150 - 5250 @ 80), (N/A, 23), (N/A), AUTO-BW
        (5250 - 5350 @ 80), (N/A, 24), (0 ms), DFS, AUTO-BW
        (5470 - 5730 @ 160), (N/A, 24), (0 ms), DFS
        (5730 - 5850 @ 80), (N/A, 30), (N/A), AUTO-BW
        (5850 - 5895 @ 40), (N/A, 27), (N/A), NO-OUTDOOR, AUTO-BW, PASSIVE-SCAN
        (5925 - 7125 @ 320), (N/A, 12), (N/A), NO-OUTDOOR, PASSIVE-SCAN
        (57240 - 71000 @ 2160), (N/A, 40), (N/A)
```

Great! We are now set to US Regulatory Domain.

Basic hostapd Configuration

Let's create a directory (Choose your location. We will use Desktop/PcapLabs) for this series.

Hostapd requires a configuration file (conf) to be passed to it at initiation (init). Let's create a basic 'hostapd' configuration file. Don't worry if you are not familiar with the config file options. We will work through them as we go through the Labs exercises.

Create a configuration file (e.g - ~/Desktop/PcapLabs/wpa2_basic.conf`).

WPA2 basic configuration.

```
interface=wlx9cefd5f6950a
driver=nl80211
ssid=PCAP-Lab
hw_mode=g
channel=6
ieee80211n=1
wmm_enabled=1

# Open network for simplest analysis (or add WPA2 below)
auth_algs=1
ignore_broadcast_ssid=0

# Optional WPA2-Personal for handshake capture
# wpa=2
# wpa_key_mgmt=WPA-PSK
# rsn_pairwise=CCMP
# wpa_passphrase=Testlab12345
```

Ok. Let's bring the interface up and start hostapd. Note: You can move to the directory where you keep your config file and ommit the full path to the confi file.

```
sudo ip link set wlx9cefd5f6950a up
sudo hostapd ~/Desktop/PcapLabs/wpa2_basic.conf
```

You should see messages indicating the interface moved to AP-ENABLED state. Verify!!

```
sudo hostapd wpa2_basic.config

wlx9cefd5f6950a: interface state UNINITIALIZED->ENABLED
wlx9cefd5f6950a: AP-ENABLED
```

Additional info can be gathered. Notice the 'type'?

```
iw dev wlx9cefd5f6950a info
```

```
Interface wlx9cefd5f6950a
        ifindex 13
        wdev 0x200000001
        addr 9c:ef:d5:f6:95:0a
        ssid PCAP-Lab
        type AP
        wiphy 2
        channel 6 (2437 MHz), width: 20 MHz, center1: 2437 MHz
        txpower 3.00 dBm
        multicast TXQ:
                qsz-byt qsz-pkt flows   drops   marks   overlmt hashcol tx-bytestx-packets
                0       0       0       0       0       0       0       0       0
```

Looking good. Now, let's create a DHCP server for our AP. Technically, this is not required for protocol analysis but it is nice to have and verify the DHCP process.

We will use a range identified by the IETF for Testing and Documentation. This will keep our lab clean as this range is reservered and non-routable.

IETF Documentation Reserved Range

```
192.0.2.0/24 is an IPv4 address range reserved specifically for
documentation, tutorials, examples, and network configuration demonstrations.

It is designated TEST-NET-1 by the IETF in RFC 5737. It is not intended for
real network communication or public Internet routing
```

Adding DHCP with dnsmasq

Clients need IP addresses. Simple dnsmasq config (dnsmasq-pcaplab.conf).

```
interface=wlx9cefd5f6950a
dhcp-range=192.0.2.10,192.0.2.20,4h
dhcp-option=3,192.0.2.1
port = 0
```

Assign an IP to the AP interface and start dnsmasq

```
sudo ip addr add 192.0.2.1/24 dev wlx9cefd5f6950a
sudo dnsmasq -C dnsmasq-pcaplab.conf
```

Verify IP assignment

```
ifconfig -a
```

```
wlx9cefd5f6950a: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.0.2.1  netmask 255.255.255.0  broadcast 0.0.0.0
        ether 9c:ef:d5:f6:95:0a  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Verify our DHCP is running

```
sudo ss -ulpn | grep ':67'
UNCONN 0      0            0.0.0.0:67        0.0.0.0:*    users:
(("dnsmasq",pid=7610,fd=4))
```

Verify Hostapd is advertising out test SSID (PCAP-Lab)

```
 BSSID              PWR  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH
ESSID

 9C:EF:D5:F6:95:0A  -43       20        0    0   6  130   WPA2 CCMP   PSK
PCAP-Lab
```

Success!!

Conclusion

We have established a working base configuration for our protocol analysis lab environment utilizing readily available software and wireless LAN adapters. We will continue to build on the base setup in each subsequent post in this series. In the next post we will look at how we can utilize Aircrack-ng and Wireshark to setup monitor interfaces and start capturing frames.